

Integral Solutions of $y^2 + x^2y + xz^2 = 2$

Complete Classification, an Infinite Family, and the Guiding Idea

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Abstract

We give an exact elementary description of every integral solution of $y^2 + x^2y + xz^2 = 2$. We then exhibit an explicit infinite family on the single fibre $x = -17$. The final section explains how the quadratic symmetry, a norm equation, and a deliberately engineered Pell unit lead naturally to the numbers occurring in that family.

1 Complete classification

Let $\{r^2 : r \in \mathbb{Z}\}$ denote the set of nonnegative integral squares.

Theorem 1 (Exact divisor–square criterion). *There is no solution with $x = 0$. For $x \neq 0$, all integral solutions are precisely*

$$(x, y, z) = \left(\varepsilon d, \quad m, \quad \delta \sqrt{-\varepsilon \left(dm + \frac{m^2 - 2}{d} \right)} \right), \quad (1)$$

where

$$\varepsilon, \delta \in \{\pm 1\}, \quad d \in \mathbb{Z}_{>0}, \quad m \in \mathbb{Z},$$

subject only to

$$d \mid m^2 - 2, \quad -\varepsilon \left(dm + \frac{m^2 - 2}{d} \right) \in \{r^2 : r \in \mathbb{Z}\}. \quad (2)$$

If the square in (2) is zero, the two choices of δ coincide; otherwise the parametrization is unique after fixing the sign of z .

Proof. If $x = 0$, the equation becomes $y^2 = 2$, impossible in $\mathbb{Z}$. Now write $x = \varepsilon d$, where $d = |x| > 0$ and $\varepsilon = \text{sgn}(x)$, and put $m = y$. Since $x^2 = d^2$, the equation is

$$m^2 + d^2m + \varepsilon dz^2 = 2.$$

Reduction modulo d gives $d \mid m^2 - 2$, and rearrangement gives

$$z^2 = -\varepsilon \left(dm + \frac{m^2 - 2}{d} \right).$$

Thus every solution has the form (1)–(2). Conversely, substituting those expressions yields

$$m^2 + d^2m + \varepsilon d \left[-\varepsilon \left(dm + \frac{m^2 - 2}{d} \right) \right] = m^2 - (m^2 - 2) = 2.$$

Hence every listed triple is a solution, and the recovery $d = |x|$, $\varepsilon = \text{sgn}(x)$, $m = y$ proves completeness and the stated uniqueness. $\square$

The classification is unconditional and effective: for any prescribed $y = m$, one checks only the finitely many positive divisors of $m^2 - 2$ and tests a single integer for being a square.

2 An explicit infinite family

The quadratic in y is best centred by

$$(2y + x^2)^2 + 4xz^2 = x^4 + 8, \quad (3)$$

which follows by multiplying the original equation by 4 and completing the square.

Proposition 2 (A Pell family on $x = -17$). *Define integer sequences (U_n) and (Z_n) by*

$$U_0 = 573, \quad Z_0 = 60, \quad \begin{cases} U_{n+1} = 33U_n + 272Z_n, \\ Z_{n+1} = 4U_n + 33Z_n \end{cases} \quad (n \geq 0). \quad (4)$$

Then, for every $n \geq 0$ and every $\sigma, \tau \in \{\pm 1\}$,

$$(x, y, z) = \left(-17, \frac{\sigma U_n - 289}{2}, \tau Z_n \right) \quad (5)$$

is an integral solution. These give infinitely many distinct solutions.

Proof. In $\mathbb{Z}[\sqrt{17}]$, recurrence (4) is exactly

$$U_{n+1} + 2Z_{n+1}\sqrt{17} = (33 + 8\sqrt{17})(U_n + 2Z_n\sqrt{17}).$$

Since $33^2 - 17 \cdot 8^2 = 1$, norms are preserved. The initial norm is

$$573^2 - 68 \cdot 60^2 = 83529 = 17^4 + 8,$$

so

$$U_n^2 - 68Z_n^2 = 17^4 + 8 \quad (6)$$

for every n . Moreover U_0 is odd and (4) preserves oddness of U_n , so the displayed y is integral. For $x = -17$, (3) is

$$(2y + 289)^2 - 68z^2 = 17^4 + 8;$$

substitution of (5) and (6) proves the claim. Finally, $U_n, Z_n > 0$ and $U_{n+1} > U_n$ by induction, so the resulting solutions are pairwise distinct as n varies. $\square$

The first two Vieta pairs are

$$\begin{aligned} &(-17, 142, \pm 60), \quad (-17, -431, \pm 60), \\ &(-17, 17470, \pm 4272), \quad (-17, -17759, \pm 4272). \end{aligned}$$

3 How the construction was found

1. Use the equation's quadratic structure. For fixed x, z , the equation is the monic quadratic

$$y^2 + x^2y + (xz^2 - 2) = 0.$$

Hence the second root corresponding to a solution y is $-x^2 - y$. This Vieta symmetry suggests translating the variable to the centre $-x^2/2$, namely setting $U = 2y + x^2$. One then obtains

$$U^2 + 4xz^2 = x^4 + 8.$$

The same equation also explains the complete classification: reducing the original equation modulo x gives $x \mid y^2 - 2$, after which solving linearly for z^2 gives Theorem 1 in one step.

2. The sign of x reveals where infinitude can occur. If $x > 0$, then $U^2 + 4xz^2 = x^4 + 8$ is positive definite; for a fixed x it admits only finitely many pairs (U, z) . Put instead $x = -a < 0$. The equation becomes

$$U^2 - a(2z)^2 = a^4 + 8, \quad (7)$$

which is a generalized Pell equation. Thus one seed solution can be multiplied repeatedly by a norm-one unit to produce infinitely many others. The task is therefore to choose a so that both the unit and the seed are simple.

3. Engineer a simple unit and a compatible seed. A particularly transparent norm-one unit appears when $a = r^2 + 1$, because

$$(2a - 1)^2 - a(2r)^2 = 1.$$

To seek a low-complexity seed for (7), take $U = 2a^2 - (r + 1)$. This automatically gives an integral $y = (U - a^2)/2$ when a and $r + 1$ have the same parity. Modulo a , the required divisibility of

$$U^2 - (a^4 + 8) = 3a^4 - 4(r + 1)a^2 + (r + 1)^2 - 8$$

reduces, using $a = r^2 + 1$, to the very small condition

$$(r + 1)^2 - 8 \equiv 2r - 8 \pmod{a}.$$

The clean choice is therefore $r = 4$, which gives

$$a = 17, \quad U = 2 \cdot 17^2 - 5 = 573.$$

The remaining coordinate is then forced:

$$z^2 = \frac{573^2 - (17^4 + 8)}{4 \cdot 17} = 3600, \quad z = 60,$$

and consequently $y = (573 - 289)/2 = 142$. Thus the seed is not a mysterious computer output: it is produced by matching a simple Pell unit with a small-offset ansatz for U .

4. Propagate the seed. For $r = 4$, the unit above is $33 + 8\sqrt{17}$. Multiplying $573 + 120\sqrt{17}$ by successive powers of this unit preserves its norm $17^4 + 8$ and gives exactly recurrence (4). Changing the sign of the square-root coefficient changes z to $-z$, while replacing U by $-U$ gives the Vieta partner $y \mapsto -17^2 - y$. These two elementary symmetries account for all four sign choices in (5).